

i When to Use

Use multiple regression to identify how multiple predictors (two or more quantitative or binary variables) contribute to explaining variation in a quantitative criterion variable. The regression model produces a linear equation and tests whether the overall model significantly predicts the criterion.

H Research Hypothesis: The Rotten Tomatoes critics score for Superman movies is significantly predicted by both the age of the actor playing Clark Kent and the year the movie was released.

H Null Hypothesis: The combination of Clark Kent actor age and year of release does not significantly predict Rotten Tomatoes critics scores for Superman movies.

Starting with Correlations

Before performing regression, examine the correlations among all variables to understand the relationships:

Correlations				
		rt_critics_score	clark_age	year
rt_critics_score	Pearson Correlation	1.000	0.128	−0.262
	Sig. (2-tailed)		.762	.530
	N	8	8	8
clark_age	Pearson Correlation	0.128	1.000	−0.234
	Sig. (2-tailed)	.762		.578
	N	8	8	8
year	Pearson Correlation	−0.262	−0.234	1.000
	Sig. (2-tailed)	.530	.578	
	N	8	8	8

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

Descriptive Statistics						
	N	Mean	Std. Deviation	Std. Error	Minimum	Maximum
rt_critics_score	8	79.38	10.53	3.72	57.00	88.00
clark_age	8	29.67	4.33	1.53	24.47	37.24
year	8	1998.12	25.30	8.95	1948.00	2025.00
Valid N (listwise)	8					

The correlation matrix helps you identify:

- Which predictors are correlated with the criterion
- Whether predictors are highly correlated with each other (multicollinearity)
- The direction and strength of relationships

All three variables should show relationships that make theoretical sense before proceeding with regression.

Note

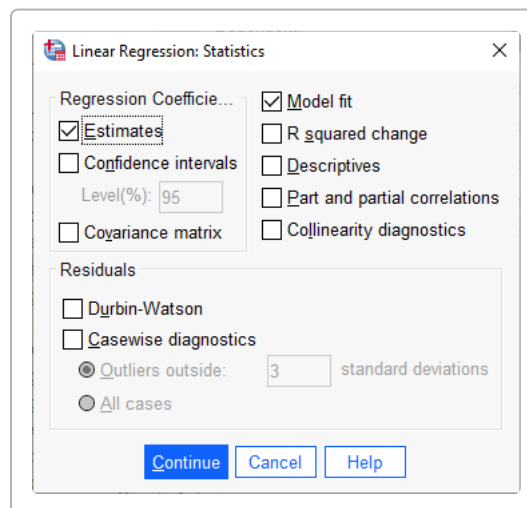
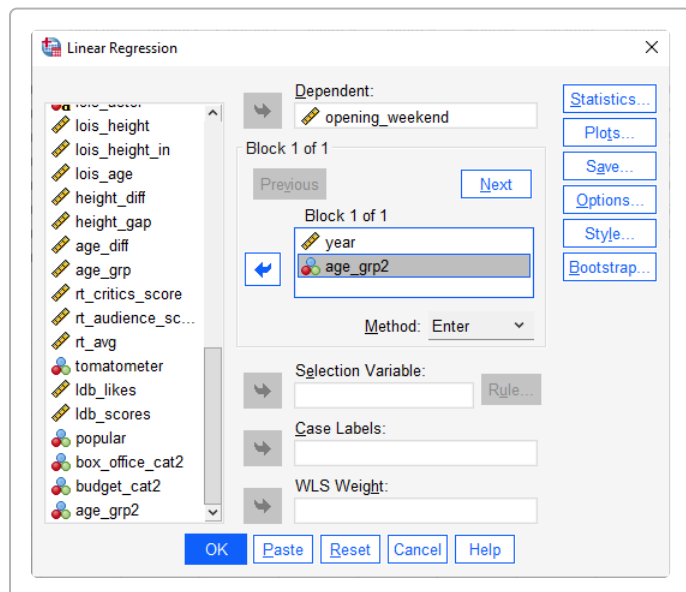
Multicollinearity: If predictors are very highly correlated with each other ($r > .80$), they may not add independent information to the regression model. Always check correlations among predictors before interpreting the regression coefficients.

Procedure: Running Multiple Regression

Analyze → Regression → Linear

In the “Statistics” window:

- Highlight the criterion variable and click the arrow to move it to “Dependent”
- Highlight each predictor variable and click the arrow to move them to “Independent(s)”
- Click “Statistics” to request additional output
- Check “Estimates” (for regression coefficients)
- Check “Confidence intervals”
- Check “Model fit” (for R^2 , adjusted R^2 , and standard error)
- Check “ANOVA table” (for F-test of overall model)
- Optionally check “Part and partial correlations”



SPSS Syntax

```
REGRESSION /STATISTICS COEFF OUTS R ANOVA /DEPENDENT rt_critics_score
/METHOD=ENTER clark_age year.
```

- 1 Request coefficients, confidence intervals, R^2 , and ANOVA table
- 2 Specify the criterion (dependent) variable
- 3 Enter all predictors simultaneously using the ENTER method

SPSS Output

Model Summary and ANOVA

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.271	.074	-.0297	11.990

ANOVA

Model	Sum of Squares	df	Mean Square	F	Sig.
Regression	57.120	2	28.560	0.199	.826
Residual	718.755	5	143.751		
Total	775.875	7			

Coefficients

Unstandardized Coefficients			Standardized Coefficients		
Model	B	Std. Error	Beta	t	Sig.
(Constant)	278.755	376.825		0.740	.493
clark age	0.172	1.076	0.071	0.160	.879

R^2 (Coefficient of Determination): The proportion of variance in the criterion explained by all predictors combined. For example, $R^2 = .35$ means that 35% of the variation in critics scores is explained by actor age and year of release.

Adjusted R^2 : A corrected version of R^2 that accounts for the number of predictors. With more predictors, adjusted R^2 decreases if those predictors don't meaningfully improve the model.

Unstandardized Coefficients			Standardized Coefficients		
Model	B	Std. Error	Beta	t	Sig.
year	−0.102	0.184	−0.246	−0.556	.602

F-test: Tests whether the overall regression model significantly predicts the criterion. A significant F-test ($p < .05$) indicates that the combination of predictors explains more variance than expected by chance.

Standard Error: The typical error in prediction. Smaller values indicate more accurate predictions.

Regression Coefficients

The coefficients table provides the unstandardized (b) and standardized (β) weights for each predictor:

- **Unstandardized coefficient (b):** The change in the criterion for a one-unit increase in that predictor, holding other predictors constant
- **Standardized coefficient (β):** Allows comparison of predictor importance on a common scale (−1 to +1)
- **t-test:** Tests whether each individual predictor significantly contributes to the model
- **Confidence Interval (CI):** The range of likely values for the true population coefficient

Interpreting Multiple Regression Results

When interpreting the coefficients:

1. **Look at the F-test first:** Does the overall model significantly predict the criterion?
2. **Check individual t-tests:** Which predictors individually contribute significantly?
3. **Compare β weights:** Larger absolute β values indicate stronger relative importance (on a standardized scale)
4. **Remember:** b weights depend on the scale of measurement, so comparing raw b values across predictors is not meaningful

Computing Predicted Scores

Once the regression equation is established, you can predict criterion scores for new cases or all cases in the dataset.

To save predicted values in SPSS:

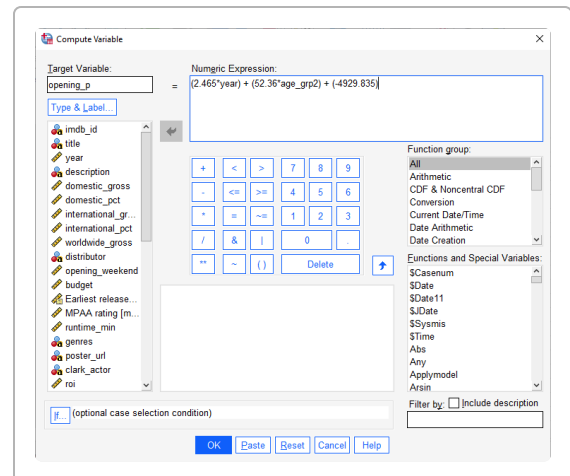
Analyze → Regression → Linear

- Complete the regression setup as before
- Click “Save...” button
- Check “Predicted values” (unstandardized)
- This creates a new variable with predicted scores

The regression equation is:

$$\text{Criterion} = a + b_1(\text{Predictor}_1) + b_2(\text{Predictor}_2) + \dots + b_k(\text{Predictor}_k)$$

Where: - a = constant (intercept) - $b_1, b_2, \dots, b_k$ = unstandardized regression coefficients from the output



Reporting the Results

Predictor	<i>B</i>	<i>SE</i>	β	<i>t</i>	<i>p</i>
Clark Age	0.17	1.08	.07	0.16	.879
Year	-0.10	0.18	-.25	-0.56	.602

Example Write-Up

Sample APA Write-Up

A multiple regression analysis was conducted to determine whether Rotten Tomatoes critics scores for Superman portrayals ($N = 8$) could be predicted from the age of the actor playing Clark Kent and the year of release. Both predictors were entered simultaneously using the ENTER method. The overall model was not significant, $F(2,5) = 0.20, p = .826$, explaining 7.4% of the variance in critics' ratings ($R^2 = 0.074$). Clark age was not a significant predictor ($b = 0.1720, Beta = 0.071, t = 0.16, p = .879$). Year was not a significant predictor ($b = -0.1023, Beta = -0.246, t = -0.56, p = .602$).

When reporting multiple regression results, be sure to communicate:

- The research hypothesis
- Sample size (N)
- The number and names of predictors
- The F -statistic, degrees of freedom, and p -value for the overall model
- The R^2 value (proportion of variance explained)
- Which predictors were individually significant ($b, Beta, t, p$)
- The direction and magnitude of significant effects
- Interpretation of what the results mean for your research question

Remember: Never report $p = .000$. Instead, report “ $p < .001$ ”.